WHYCODE

Why is this code like this? Ask its own history.

Team FORTEC · hackFront India 2026 · AI and Developer Tools (T-1)

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Built on django/django

THE PROBLEM

- Codebases accumulate decisions **nobody can explain** any more
- A new developer reads an odd function and **guesses** why it exists
- A wrong guess breaks something the original author **fixed on purpose**
- The reasons live scattered across **commits, pull requests and tickets** — never in one place, never linked to the code they explain

THE SOLUTION: WHYCODE

- **Ingest** a public repo's history into a graph
- **Ask** plain-language *why* questions
- Get a **cited evidence chain** and a confidence score
- **Refuses** when no evidence exists — no invented answers, ever

HOW IT WORKS

1. **Ingest** — gh CLI for commits and PRs, plus Trac tickets, since Django uses Trac, not GitHub Issues
2. **Graph** — entities: commit, PR, ticket, file, person; edges: authored, fixes, references, touches, supersedes
3. **Ask** — evidence chain, supersession flags, and refusal when evidence is absent
4. **Runs anywhere** — the bundled SQLite database needs no credentials

DEMO: A REAL ANSWER FROM DJANGO'S HISTORY

```
$ python -m whycode.cli --question \
    "Why does ignore_warnings match substrings by default?"

refused: false   confidence: 0.99

Best match: Fixed #37089 — Made ignore_warnings()
match substrings by default. (commit)

Evidence:
→ fixes      Trac ticket #37089
→ authored by HelmiDev03
```

DEMO: THE TEMPLATE-TAGS OPTION

```
$ python -m whycode.cli --question \
    "Why is there a transitional option to disable
     multiline template tags?"

refused: false   confidence: 0.86

Best match: Refs #35899 — Added allow_multiline_tags
transitional template option. (commit)

Evidence:
→ references Trac ticket #35899
→ authored by boxed
```

THE REFUSAL, HONESTLY DEMONSTRATED

```
$ python -m whycode.cli --question \
    "Why does Django use a microservices architecture?"

refused: true   confidence: 0.0

No evidence in the graph answers this.
Refusing to invent an answer.
```

> An earlier version answered this from an unrelated docs commit; the absent-term rule + regression test fixed it.

ENGINEERING: RUNNABLE, TESTED, NO MAGIC

200	100	100	161	90
commits	pull requests	Trac tickets	files	people

630
edges

Python standard library only · SQLite graph · **10/10 tests** · bundled database runs with zero credentials

WHYCODE — BY FORTEC

Repo: github.com/eres0045/hackfront-fortec

```
$ python -m whycode.cli --db data/whycode.db \
    --question "Why ...?"
```

Questions welcome.